The Lagrange points: the key equations

Equation numbers are the astronomy handout's (Section 1). Everything else is yours to fill in.

Set-up (Steps 1–3). Two primaries M_1, M_2 a fixed distance R apart, circling their CM at angular speed Ω; a test mass $m \ll M_1, M_2$ moving in their plane ($Z = 0$). With $\mu \equiv M_2/(M_1 + M_2)$:

$$r_1 = \mu R, \qquad r_2 = (1-\mu)R \quad \text{(distances of } M_1, M_2 \text{ from the CM)}, \qquad \Omega^2 = \frac{G(M_1+M_2)}{R^3} \quad (1.4).$$

Step 4: coordinates that rotate with the primaries. (X, Y) inertial, (x, y) turned through Ωt; the primaries then sit at $(-r_1, 0)$ and $(r_2, 0)$ for all time, so U loses its t:

$$X = x\cos\Omega t - y\sin\Omega t, \qquad Y = x\sin\Omega t + y\cos\Omega t \quad (1.7),$$

$$U(x,y) = -\frac{GmM_1}{\sqrt{(x+r_1)^2+y^2}} - \frac{GmM_2}{\sqrt{(x-r_2)^2+y^2}} \quad (1.9),$$

$$\dot{X}^2 + \dot{Y}^2 = \dot{x}^2 + \dot{y}^2 + 2\Omega(x\dot{y} - y\dot{x}) + \Omega^2(x^2+y^2),$$

$$\boxed{\mathcal{L} = \tfrac{1}{2}m(\dot{x}^2+\dot{y}^2) + m\Omega\,(x\dot{y} - y\dot{x}) + \tfrac{1}{2}m\Omega^2(x^2+y^2) - U(x,y)} \quad (1.10)\text{: no } t \text{ in it.}$$

Step 5: Lagrange's equations. $\dfrac{d}{dt}\dfrac{\partial \mathcal{L}}{\partial \dot{x}} = \dfrac{\partial \mathcal{L}}{\partial x}$ and the same for y give

$$m\ddot{x} = -\frac{\partial U}{\partial x} + m\Omega^2 x + 2m\Omega\dot{y}, \qquad m\ddot{y} = -\frac{\partial U}{\partial y} + m\Omega^2 y - 2m\Omega\dot{x} \quad (1.11),$$

Newton's second law with three forces: gravity $-\boldsymbol{\nabla}U$; centrifugal $m\Omega^2(x, y)$, straight away from the CM; Coriolis $2m\Omega(\dot{y}, -\dot{x})$, perpendicular to the velocity, no work done.

Step 6(a): where can m sit still? $\dot{x} = \dot{y} = \ddot{x} = \ddot{y} = 0$ in (1.11): the Coriolis terms vanish with the velocities, and

$$\boldsymbol{\nabla}U_{\text{eff}} = 0, \qquad U_{\text{eff}} \equiv U - \tfrac{1}{2}m\Omega^2(x^2+y^2) \quad (1.12).$$

Exactly five solutions. On the axis ($y = 0$), in units $R = 1$, $G(M_1 + M_2) = 1$, per unit mass:

$$x = \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} + \frac{\mu\,(x-1+\mu)}{|x-1+\mu|^3} \quad (1.17): \quad L_1, L_2, L_3, \text{ one in each stretch of the axis;}$$

for small μ, L_1 and L_2 sit $d \approx R\,(\mu/3)^{1/3}$ from M_2 (1.18), the Hill radius. Off the axis, L_4 and L_5 at the two equilateral points ($d_1 = d_2 = R$).

Step 6(b): what is conserved? No t in $\mathcal{L}$, so

$$\mathcal{H} = \tfrac{1}{2}m(\dot{x}^2+\dot{y}^2) + U_{\text{eff}} \quad (1.15), \text{ the Jacobi integral; not } T + U.$$

Stability. None of the five is a minimum of U_{eff} (L_1–L_3 saddles, L_4, L_5 maxima). Linearize (1.11) about a point with the Coriolis term kept ($x = L_x + \xi$, $y = L_y + \eta$; A, B, C the second derivatives of U_{eff} there; same units as above):

$$\ddot{\xi} - 2\dot{\eta} = -A\xi - B\eta, \qquad \ddot{\eta} + 2\dot{\xi} = -B\xi - C\eta \quad (1.25);$$

$$\xi, \eta \propto e^{\lambda t}: \qquad \lambda^4 + (4 + A + C)\lambda^2 + (AC - B^2) = 0 \quad (1.27).$$

The 4 is the Coriolis effect. A saddle ($AC - B^2 < 0$) always has a real $\lambda > 0$: L_1–L_3 unstable for every μ (Sun–Earth L_2: e-folding time 23 days; JWST fires thrusters every three weeks). At L_4, $A = -\frac{3}{4}$, $C = -\frac{9}{4}$, $AC - B^2 = \frac{27}{4}\mu(1-\mu)$:

$$\lambda^4 + \lambda^2 + \tfrac{27}{4}\mu(1-\mu) = 0 \quad \Rightarrow \quad \text{stable iff } 27\mu(1-\mu) < 1, \;\; \mu < 0.0385 \quad (1.37),$$

which covers Sun–Jupiter (the Trojans; libration period 147 yr) and Earth–Moon.

	μ	L_1 from M_2	L_2 from M_2	$R(\mu/3)^{1/3}$
Sun–Earth	3.00×10^{-6}	1.49×10^6 km	1.50×10^6 km	1.50×10^6 km
Sun–Jupiter	9.54×10^{-4}	5.19×10^7 km	5.43×10^7 km	5.31×10^7 km
Earth–Moon	1.22×10^{-2}	5.80×10^4 km	6.45×10^4 km	6.13×10^4 km